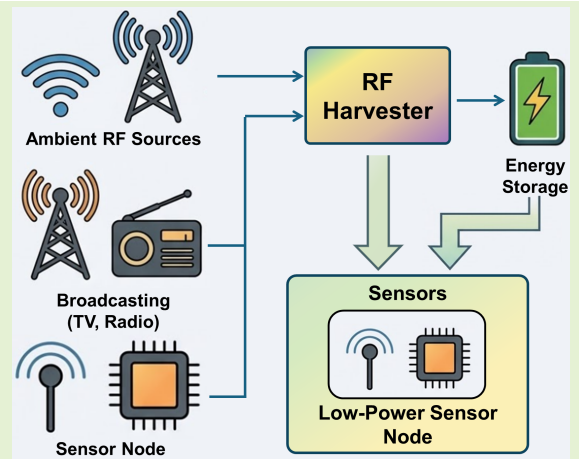


RF Energy Harvesting Integrated Circuits for Low-Power Sensors: A Review

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Abstract—With the rapid proliferation of the Internet of Things (IoT), billions of distributed sensors have been deployed worldwide, creating overwhelming challenges associated with battery replacement and perpetual maintenance. To address these critical energy bottlenecks, Radio Frequency Energy Harvesting (RFEH) has emerged as a transformative solution for enabling battery-less, self-sustaining sensor nodes. In this manuscript, we review the design challenges and recent advances in RFEH front ends tailored for such low-power sensing systems. Specifically, the fundamental limitations imposed by extremely low ambient RF energy density and nonlinear rectifier impedance are analyzed. To establish the circuit-level foundation of RFEH operation, the Cross-Coupled Differential-Drive (CCDD) rectifier is adopted as a representative topology to analyze low-power conduction, cycle-by-cycle charge balance, and reverse-leakage mechanisms. Various threshold-voltage compensation strategies, leakage-suppression techniques, and tunable matching networks are surveyed to highlight trade-offs among sensitivity, efficiency, and area. Building on this rectifier-level foundation, the discussion is extended to system-level RFEH design, including multiband energy aggregation, wide-input-range operation, interactions with power-management units, energy storage and regulation, and load-aware energy delivery for stable long-term sensor operation. Finally, future research directions and a prospective architecture for enhanced RFEH efficiency are presented.

Index Terms—Battery-less, maintenance-free, RF energy harvesting, power management, low-power, sensor nodes, silicon.



I. INTRODUCTION

THE Internet of Things (IoT) has brought convenience to modern society, but it has also led to the proliferation of billions of sensor devices and an increasing demand for energy [1], [11]. To ensure longer standby and monitoring periods for these devices, low-power sensors have become a major research focus over the past several decades. Advances in process nodes and the development of low-power technologies have driven the power consumption of low-frequency sensors to the microwatt (μW) and even nanowatt (nW) levels [2]–[4]. However, these small sensors still heavily rely on chemical batteries and face critical challenges, including leakage risks, high replacement costs, environmental contamination, and

form-factor constraints [25]. These issues limit the potential applications of these small sensors and result in high maintenance costs.

To address these challenges and further improve the standby time of low-power devices while reducing maintenance costs, self-powered, battery-less systems based on energy harvesting technologies have become a key area of research in recent years. These systems aim to convert various potential environmental energy sources, such as solar, wind, thermal, and radio frequency (RF) energy into electrical energy that can be directly utilized by integrated circuit systems. This approach eliminates battery dependency and allows ultra-long standby times without the need for continuous maintenance [8]–[10]. Among these environmental energy sources, RF energy harvesting (RFEH) has gained widespread application in battery-free systems for low-power devices due to its stability and weather independence [11], [53], [54].

However, as illustrated in Fig. 1, the low ambient RF energy density remains a major challenge for RFEH systems [17], [19]. Such weak signals often fail to overcome the threshold voltage of transistors in standard CMOS processes, thereby limiting the rectification efficiency under low-input-

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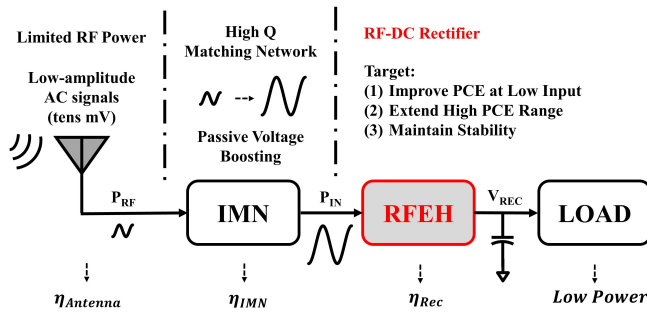


Fig. 1. Overview of a typical RF Energy Harvesting (RFEH) system.

power conditions and making it difficult to generate a sufficient output voltage. Although impedance matching networks can enhance the amplitude of the RF signal received by the rectifier, their dependence on high-Q inductors constrains the achievable gain while consuming considerable on-chip area and increasing fabrication cost [25]. To improve the efficiency under low-power conditions and enhance the overall sensitivity, many studies have proposed threshold-voltage compensation techniques to overcome intrinsic device limitations. Although both Dickson-type and cross-coupled differential-drive (CCDD) rectifiers have been widely employed in CMOS RFEH systems, the accumulated threshold-voltage loss of cascaded Dickson stages limits their sensitivity under weak RF excitation. Consequently, CCDD rectifiers have become particularly prominent in sensor-oriented far-field RFEH systems because their differential cross-coupled gate drive dynamically alleviates the effective threshold-voltage penalty [26], [41], [69], [70], [82]. However, the bidirectional conduction inherent in the switch-based CCDD structure can also cause reverse charge flow as the input power and rectified output voltage increase.

As the incident RF power increases, the rectified output voltage may exceed the instantaneous RF input voltage during part of each cycle, causing charge stored at the output to flow back toward the RF source [39], [43], [44]. This reverse leakage reduces the net charge delivered to the load and results in a rapid degradation of the power conversion efficiency (PCE). Such performance deterioration is detrimental to the feasibility of practical applications. Consequently, implementing leakage suppression mechanisms and maintaining high PCE over a wide input dynamic range have emerged as pivotal research objectives.

Driven by these challenges, numerous rectifier architectures and system-level optimization strategies have been proposed. Recent surveys have reviewed application-oriented rectenna design, 5G/mmWave systems, rectenna arrays, and rectifier operation across frequency, bandwidth, and input-power conditions [89]–[93]. More closely related to integrated RFEH front ends, multiband architectures, wide-PDR CMOS rectifiers, and reverse-leakage mitigation have also been systematically summarized [41], [82], [94]. Complementing these studies, this review adopts a bottom-up, device-to-system perspective grounded in a CCDD-based analytical framework. The detailed CCDD analysis links device-level subthreshold

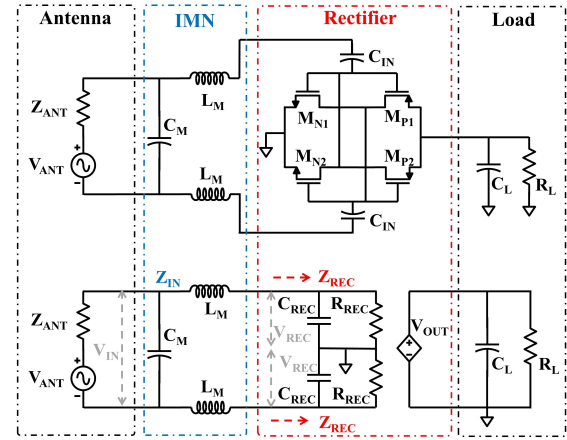


Fig. 2. Circuit and equivalent electrical model of the RFEH front-end based on Cross-Coupled Differential-Drive (CCDD) rectifier.

behavior and cycle-by-cycle charge balance to threshold-voltage compensation and reverse-leakage suppression, while the subsequent system-level discussion extends these circuit-level insights to multiband harvesting, wide-PDR operation, PMU interaction, energy storage and delivery, and long-term stable sensor operation.

The remainder of this paper is organized as follows. Section II introduces the RFEH front-end architecture and develops the analytical framework for efficiency, sensitivity, and charge balance. Section III reviews threshold-voltage-compensation and leakage-suppression techniques and compares their principal tradeoffs. Section IV presents application-oriented design guidelines, practical implementation challenges, system-level strategies for stable long-term operation, and future research directions. Section V concludes the paper.

II. OVERVIEW OF RFEH FRONT-END

Fig. 2 illustrates the circuit architecture of an RFEH front end based on a CCDD rectifier, along with its equivalent circuit model. The antenna captures ambient RF energy, while the impedance matching network minimizes reflection loss at the target frequency and passively boosts the signal amplitude. The CCDD rectifier then performs AC-to-DC conversion, generating a DC output voltage to power the load. In many practical applications, a balun is also commonly incorporated into an RFEH system to enable single-ended to differential signal conversion.

The following analysis uses the CCDD rectifier as a common circuit-level framework because it captures both threshold-limited forward conduction under weak RF excitation and reverse charge flow at elevated input power. This framework connects sensitivity, threshold-voltage compensation, leakage suppression, and wide-PDR operation, while alternative rectifier and system architectures are discussed subsequently according to their distinct operating mechanisms and design tradeoffs.

A. Impedance Matching Network

Since RF signals used as energy sources typically span a wide frequency range from 3 kHz to 300 GHz [11], directly

connecting the antenna to the rectifier can result in severe reflection losses, significantly limiting the input power (P_{in}) received by the RFEH front end. Consequently, the rectifier often fails to generate sufficient output voltage under low-input conditions. Therefore, inserting an impedance matching network between the antenna and the rectifier is an indispensable part of the system design [12]–[14].

However, the design of the matching network presents several challenges. Because the rectifier is a highly nonlinear circuit, its input impedance varies dynamically with both the input power and the operating frequency. As a result, conventional fixed passive matching networks (such as L-type or π -type configurations) can only achieve optimal matching at a specific frequency and input power level, where the reflection coefficient (S_{11}) is minimized. Given that RF energy sources usually exhibit very low input power, designers typically optimize the matching point at low power levels to achieve higher matching efficiency under weak-signal conditions, thereby improving the overall PCE and sensitivity. As the input power increases, however, the rectifier input impedance decreases, causing the efficiency of the fixed matching network to degrade rapidly, which in turn leads to a decline in the overall system efficiency. To maintain high-quality matching over a wider input power range, several studies have proposed tunable or reconfigurable matching networks [15], [16], which adaptively adjust the matching network according to output conditions, or have employed alternative network topologies [14]. Other approaches modify the rectifier itself, utilizing reconfigurable architectures to dynamically adjust its input impedance and thereby achieve broadband impedance matching [17], [18], [47].

In addition, to capture more ambient energy and operate robustly in complex environments, some RFEH systems are designed to function across multiple frequency bands [20]–[23]. To ensure effective matching across these bands, various multi-band matching networks have been developed, including those based on switchable topologies or multi-resonant networks, which enable band-selective impedance matching and enhanced energy harvesting performance.

Besides reducing reflection loss, another critical function of the impedance matching network in RFEH applications is to amplify the amplitude of the weak input AC signal. This amplification allows the RFEH system to overcome the threshold voltage for rectification even under lower input conditions, thereby achieving higher sensitivity and efficiency. The work in [25] provides a detailed derivation of this. Taking the equivalent circuit model and the L-type matching network illustrated in Fig. 2 as an example, the antenna can be modeled as an AC voltage source V_{ANT} with an impedance Z_{ANT} . Consequently, the energy available at the antenna is given by [24]:

$$P_{RF} = P_{ANT} = \left(\frac{V_{ANT}}{\sqrt{2}} \right)^2 \left| \frac{Z_{IN}}{Z_{IN} + Z_{ANT}} \right|^2 \Re \left\{ \frac{1}{Z_{IN}} \right\}, \quad (1)$$

where Z_{IN} denotes the input impedance of the matching network. For an ideal matching network, the imaginary part of Z_{ANT} is canceled out by that of Z_{IN} , while their real parts

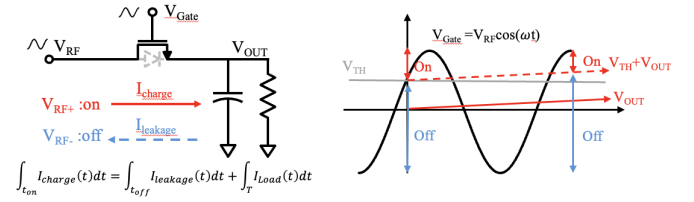


Fig. 3. Principle of RFEH front-end.

are equal. Consequently, the amplitude of the AC signal at the antenna input can be expressed as:

$$V_{ANT} = \sqrt{8 \times R_{ANT} \times P_{RF}}. \quad (2)$$

Considering that the antenna resistance R_{ANT} typically has a value of 50Ω and the input power of ambient RF energy generally ranges from -30 dBm to -20 dBm, substituting these values into the equation reveals that the amplitude of the AC signal at the antenna is merely between 20 mV and 63 mV. Such a small voltage amplitude is insufficient to drive the rectifier into normal operation. To obtain a first-order voltage-transformation estimate, a fixed L-type matching network is considered at a selected nominal operating point. At this point, the nonlinear rectifier is represented by its fundamental-frequency differential input impedance, and the matching network is assumed to be narrowband, lossless, and conjugately matched, with the rectifier input reactance fully canceled. Under these assumptions, the fundamental input power equals the total power delivered to the two symmetric rectifier branches, giving

$$\frac{V_{IN}^2}{R_{IN}} = 2 \frac{V_{REC}^2}{R_{REC}} \quad (3)$$

where $V_{IN} = V_{ANT}/2$, and R_{REC} denotes the equivalent input resistance of the rectifier. Therefore, at the selected matching point, the single-ended voltage boosting ratio of each rectifier input branch, denoted by $A_{V,boost}$, can be approximated as

$$A_{V,boost} = \frac{V_{REC}}{V_{ANT}/2} = \sqrt{\frac{R_{REC}}{2R_{ANT}}} = \frac{\sqrt{Q^2 + 1}}{2} \quad (4)$$

$$\simeq \frac{Q}{2} \quad (R_{REC} \gg R_{ANT}).$$

Here, $Q = \sqrt{2R_{REC}/R_{ANT} - 1}$ denotes the impedance-transformation quality factor defined using the differential rectifier input resistance $R_{REC,diff} = 2R_{REC}$, while $A_{V,boost}$ represents the single-ended voltage gain of each rectifier input branch. Under the ideal matching assumptions, (4) provides a first-order estimate of the voltage transformation achievable at the selected operating point, illustrating how passive impedance transformation can help the rectifier overcome the threshold-voltage limitation under weak-input conditions.

However, this relation should not be interpreted as a general voltage-boost law for nonlinear rectifiers. As the input power, operating frequency, output voltage, or load condition changes, the rectifier input impedance deviates from the value used to design the fixed matching network, causing $|S_{11}|$ to increase and reducing the power delivered to the rectifier. Consequently,

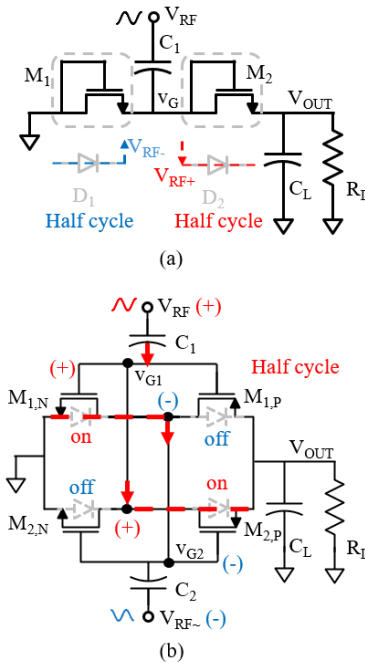


Fig. 4. Rectifier circuits: (a) A single-stage Dickson rectifier, (b) a single-stage CCDD rectifier.

the actual V_{REC} may increase much more slowly than predicted by (4), while finite component quality factors, parasitic losses, and harmonic generation further reduce the achievable voltage transformation. This operating-point-dependent mismatch also contributes to the limited PDR of fixed-matching-network RFEH systems.

B. Model and analysis of CCDD rectifier under low RF-input

Fig. 3 illustrates the basic principle of the RFEH front-end. To better integrate with the rest of the system-on-chip (SoC), designers often replace traditional diodes with CMOS transistors for rectifying the input RF signals. In order to synchronize the transistor switch with the input RF signal without requiring additional biasing, the gate of the MOS transistor is driven by a voltage derived from the differential RF input. When the V_{RF} input signal exceeds the output DC node voltage, the transistor turns on to charge the capacitor, while it remains off during other conditions to minimize leakage from the output V_{OUT} node to the V_{RF} input node. Each cycle, the RF signal charges the output node until a balance between charging and discharging is achieved for each period, as expressed in Equation (5).

$$\int_{t_{on}} I_{charge}(t) dt = \int_{t_{off}} I_{leakage}(t) dt + \int_T I_{Load}(t) dt, \quad (5)$$

Based on this fundamental principle, RFEH front-ends can be categorized into two basic architectures: the Dickson rectifier and the CCDD rectifier, as shown in Fig. 4.

The Dickson rectifier typically uses single-ended input and performs rectification through diode-connected MOS transistors. During the negative half-cycle of the RF signal (V_{RF}),

transistor D_1 turns on, allowing the coupling capacitor C_1 to charge and accumulate energy. During the positive half-cycle, D_2 turns on, transferring the accumulated charge from C_1 along with the input charge to the load storage capacitor C_L . As a result, the V_{RF} signal needs to overcome the threshold voltage (V_{TH}) during both positive and negative cycles.

Unlike the Dickson rectifier, the CCDD rectifier adopts a differential input structure, with the charge accumulation process occurring during the opposite half-cycle of the previous period. Taking a positive V_{RF+} as an example, transistor $M_{1,N}$ turns on, charging the coupling capacitor C_2 . At the same time, $M_{2,P}$ also turns on, allowing the charge accumulated in C_1 during the negative half-cycle of the previous period (via $M_{2,N}$) to charge the load capacitor C_L in the current cycle. In the next half-cycle, the charge stored in C_2 is transferred to C_L as $M_{1,P}$ turns on, while C_1 begins recharging via $M_{2,N}$ in preparation for the next cycle. This alternating process continues, enabling high-efficiency energy harvesting.

Due to the complementary transistor structure of the CCDD rectifier, the positive and negative half-cycles of the input AC signal simultaneously contribute to transistor conduction, inherently providing dynamic compensation. This mechanism enables the rectifier to overcome the threshold voltage limitations under low-input conditions, thereby underscoring its widespread adoption in RFEH applications. However, even with such inherent compensation, the CCDD rectifier typically operates in the subthreshold region during the turn-on phase, where its charging current can be expressed by (6):

$$I_{Dsub} = I_S \frac{W}{L} e^{\frac{V_{GS}}{nV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}} \right) (1 + \lambda_{sub} V_{DS}), \quad (6)$$

where I_S is a function of some process-related constants. I_S is exponentially dependent on the threshold voltage of the transistor, which takes the form

$$I_S = I_{S0} e^{\frac{-V_{TH}}{nV_T}}, \quad (7)$$

Therefore, by analyzing the gate-source and source-drain voltages of the transistor, we can derive the current expressions for both the forward charging and reverse leakage stages. Assuming that the rectifier has achieved dynamic charge balance, the charge balance expression is given as follows:

$$\Delta Q_{sub,charge} = \Delta Q_{sub,reverse} + \Delta Q_{Load}, \quad (8)$$

At this point, the charging node v_{G1} and v_{G2} of the CCDD rectifier can be considered as the sum of a DC component V_G and the input RF AC component V_{RF} :

$$v_{G1} = V_G + v_{RF} = \frac{V_{OUT}}{2} + v_{RF}, \quad (9)$$

$$v_{G2} = V_G - v_{RF} = \frac{V_{OUT}}{2} - v_{RF}, \quad (10)$$

The source-drain voltage and the gate-source voltage of transistor $M_{1,N}$ and $M_{2,P}$ can be expressed as:

$$V_{GS,M_{1,N}} = v_{G1} - 0 = V_G + V_{RF} \cdot \sin \omega t, \quad (11)$$

$$V_{DS,M_{1,N}} = v_{G2} - 0 = V_G - V_{RF} \cdot \sin \omega t, \quad (12)$$

$$V_{SG,M_{2,P}} = V_{OUT} - v_{G2} = (V_{OUT} - V_G) + V_{RF} \cdot \sin \omega t, \quad (13)$$

$$V_{SD,M2,P} = V_{OUT} - v_{G1} = (V_{OUT} - V_G) - V_{RF} \cdot \sin \omega t, \quad (14)$$

It is worth noting that, because the source and drain terminals of CMOS transistors are physically symmetric, some designers adopt an alternative modeling approach [26], [27]. Taking $M_{1,N}$ as an example, the node connected to v_{G2} is treated as the source, while the node connected to ground is treated as the drain, resulting in a gate-source voltage expressed as $V_{GS} = v_{G1} - v_{G2}$. This representation clearly illustrates the V_{GS} enhancement and compensation effect introduced by the CCDD structure. However, many designers neglect the impact of V_{SB} during modeling. In reality, the dynamically varying V_{SB} introduces a non-negligible influence on the threshold voltage V_{th} , as shown below:

$$V_{th} = V_{th0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right). \quad (15)$$

This increases the complexity of both modeling and simulation, and therefore the first modeling approach is often preferred in practice. The steady-state V_{GS} waveforms of $M_{1,N}$, $M_{2,N}$, $M_{1,P}$ and $M_{2,P}$ are shown in Fig. 5. The DC bias provides compensation for both the NMOS and PMOS devices, enabling them to more easily overcome the threshold voltage limitation during the turn-on phase.

By combining (11), (12), (13), (14) and (6), while considering that the value of n in (6) is typically approximated as 1, we can derive and simplify the forward and reverse charging current expressions for $M_{1,N}$ and $M_{2,P}$ as follows:

$$I_{Dsub,M1,N} = I_{S,N} \frac{W_N}{L_N} \left(e^{\frac{V_G + V_{RF} \cdot \sin \omega t}{V_T}} - e^{\frac{2V_{RF}}{V_T}} \right) \times (1 + \lambda_{sub}(V_G - V_{RF} \cdot \sin \omega t)). \quad (16)$$

$$I_{Dsub,M2,P} = I_{S,P} \frac{W_P}{L_P} \left(e^{\frac{(V_{OUT} - V_G) + V_{RF} \cdot \sin \omega t}{V_T}} - e^{-\frac{2V_{RF}}{V_T}} \right) \times (1 + \lambda_{sub}((V_{OUT} - V_G) - V_{RF} \cdot \sin \omega t)). \quad (17)$$

Due to the inherent symmetry of the CCDD architecture, $M_{1,N} = M_{2,N}$ and $M_{1,P} = M_{2,P}$, the two pairs of CMOS transistors ideally contribute equally to the output within each cycle. The NMOS devices accumulate charge onto the input capacitors C_1 or C_2 , while the PMOS devices extract this charge and deliver it to the load capacitor, generating the output voltage V_{out} . Under steady-state conditions, the net charge stored on the input capacitors averages to zero, indicating that the NMOS and PMOS transistors contribute an equal amount of charge over a single cycle. Consequently, although the NMOS transistors do not directly deliver charge to the load capacitor C_L , the total charge transferred to C_L by the two PMOS devices within one cycle can be expressed as:

$$\Delta(Q_{sub,fw} + Q_{sub,re})_{M1,N} + \Delta(Q_{sub,fw} + Q_{sub,re})_{M2,P} = 2 \cdot \Delta(Q_{sub,fw} + Q_{sub,re})_{M1,P} = \Delta Q_{Load}, \quad (18)$$

Combining Fig. 5, the transistor pairs $M1,N$ and $M2,P$ can be extracted for separate analysis. Each device is ON during $t \in [t_1, T/2 - t_1]$ (or $t \in [t_2, T/2 - t_2]$ for the second pair), providing forward charging currents, and operates in a non-ideal OFF interval $t \in [T/2 - t_1, T + t_1]$ (or $t \in [T/2 - t_2, T + t_2]$) where reverse leakage currents appear.

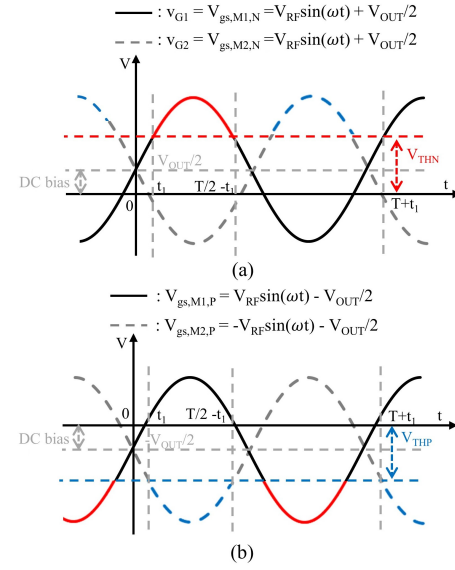


Fig. 5. Voltage waveforms of the single-stage CCDD rectifier circuit.

Expanding and simplifying equation (18) therefore gives the charge-balance relation over one period as

$$\begin{aligned} & \left(\int_{t_1}^{T/2 - t_1} I_{Dsub,M1,N,fw}(t) dt + \int_{T/2 - t_1}^{T + t_1} I_{Dsub,M1,N,re}(t) dt \right) \\ &= \left(\int_{t_2}^{T/2 - t_2} I_{Dsub,M2,P,fw}(t) dt + \int_{T/2 - t_2}^{T + t_2} I_{Dsub,M2,P,re}(t) dt \right) \\ &= \int_0^T \frac{1}{2} I_{Load}(t) dt, \end{aligned} \quad (19)$$

Therefore, by substituting (16) and (17) into (19), neglecting channel-length modulation by setting $\lambda_{sub} = 0$, we obtain the following equation for a single cycle of the single-stage CCDD rectifier under steady-state equilibrium:

$$\begin{aligned} & \int_0^T K_N e^{\frac{V_G + V_{RF} \cdot \sin \omega t}{V_T}} dt - \int_0^T K_N e^{\frac{2V_{RF}}{V_T}} dt \\ &= \int_0^T K_P e^{\frac{(V_{OUT} - V_G) + V_{RF} \cdot \sin \omega t}{V_T}} dt - \int_0^T K_P e^{-\frac{2V_{RF}}{V_T}} dt \\ &= \frac{1}{2} \int_0^T I_{Load} dt \end{aligned} \quad (20)$$

where K_N and K_P represent $I_{S,N} \cdot W_N/L_N$ and $I_{S,P} \cdot W_P/L_P$, respectively, I_{Load} denotes the load current of the single-stage CCDD rectifier, and T represents one period after the output has stabilized. Using the integral representation of the zeroth-order modified Bessel function of the first kind, equation (20) can be simplified as

$$\begin{aligned} & K_N e^{\frac{V_G}{V_T}} \cdot I_0\left(\frac{V_{RF}}{V_T}\right) - K_N \cdot I_0\left(\frac{2V_{RF}}{V_T}\right) \\ &= K_P e^{\frac{V_{OUT} - V_G}{V_T}} \cdot I_0\left(\frac{V_{RF}}{V_T}\right) - K_P \cdot I_0\left(\frac{2V_{RF}}{V_T}\right) \\ &= \frac{1}{2} I_{Load} \end{aligned} \quad (21)$$

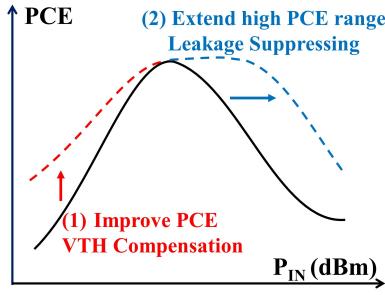


Fig. 6. Benefits of threshold-voltage compensation and leakage suppression.

where $I_0(x)$ is expressed as

$$I_0(x) = \frac{\omega}{2\pi} \int_0^T e^{x \sin(\omega t)} dt \quad (22)$$

Introducing the modified Bessel function $I_0(x)$ resolves the integration of exponential sinusoidal terms, effectively converting the AC dynamics into a closed-form DC steady-state equation. By simplifying equation (21), we obtain:

$$V_G = V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_N \cdot I_0\left(\frac{2V_{RF}}{V_T}\right)}{K_N \cdot I_0\left(\frac{V_{RF}}{V_T}\right)} \right). \quad (23)$$

$$V_{OUT} = V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_N \cdot I_0\left(\frac{2V_{RF}}{V_T}\right)}{K_N \cdot I_0\left(\frac{V_{RF}}{V_T}\right)} \right) + V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_P \cdot I_0\left(\frac{2V_{RF}}{V_T}\right)}{K_P \cdot I_0\left(\frac{V_{RF}}{V_T}\right)} \right). \quad (24)$$

In the absence of additional bias compensation, an appropriate design of the NMOS and PMOS width-length ratios allows us to set $K_N = K_P = K$. Since electron mobility is higher than hole mobility ($\mu_n > \mu_p$), the PMOS aspect ratio (W_P/L_P) is generally increased to balance the effective NMOS and PMOS drive strengths, thereby obtaining $K_N \approx K_P$ and ensuring symmetric charge transfer. This symmetric sizing further simplifies equations (23) and (24) as follows:

$$V_{OUT} = 2 \cdot V_G = 2 \cdot V_T \ln \left(\frac{\frac{I_{Load}}{2 \cdot K} + I_0\left(\frac{2V_{RF}}{V_T}\right)}{I_0\left(\frac{V_{RF}}{V_T}\right)} \right). \quad (25)$$

Based on this output voltage, the efficiency can be calculated using the following formula:

$$PCE = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT}^2 / R_{Load}}{P_{IN}}. \quad (26)$$

III. COMPENSATION AND LEAKAGE CONTROL TOPOLOGIES

The derivations in Section II reveal that the output voltage of the CCDD rectifier is established by the dynamic equilibrium among the forward charging current, reverse leakage current, and load current. Under extremely weak RF inputs, the charging current becomes insufficient to overcome the intrinsic threshold voltage of MOS transistors, preventing the rectifier from producing a usable output voltage. Consequently, improving the effective transistor conduction capability becomes

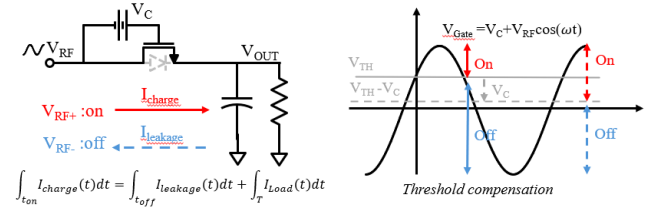


Fig. 7. Principle of compensation for threshold voltage.

the primary design objective, motivating the development of various threshold-voltage compensation techniques.

However, as the input power further increases, the forward charging current is no longer the dominant performance bottleneck. Instead, the reverse leakage current increases rapidly with the rising output voltage and can progressively become a dominant loss mechanism in CCDD rectifiers, leading to degraded PCE and a reduced power dynamic range (PDR). Therefore, the optimization focus naturally shifts from enhancing forward conduction under ultra-low input conditions to suppressing reverse leakage under moderate and high input conditions, as illustrated in Fig. 6.

Table I provides a comparison between threshold-voltage compensation and leakage suppression techniques. These two techniques target different operating regimes and address complementary performance limitations of RF energy harvesters. Accordingly, this section reviews threshold-voltage compensation and leakage-suppression techniques from this unified design perspective to achieve both ultra-high sensitivity and high-efficiency operation over a wide input-power range.

A. VTH Compensation Topologies

Threshold-voltage compensation techniques were initially widely employed in Dickson rectifiers. As shown in Fig. 7, a common approach is to apply an additional DC bias voltage V_C directly to the transistor gates, enabling the RF signal to overcome the threshold voltage at a much lower amplitude and charge the load capacitor, thereby achieving a higher output voltage [28]–[30]. In addition to this direct method, some designers adopt indirect compensation strategies. For example, modifying the substrate bias to influence V_{SB} in equation (15), thereby reducing the effective threshold voltage. The compensated output voltage of a single-stage Dickson rectifier can be expressed as [31]:

$$V_{o1} = nV_T \ln \left(\frac{I_0\left(\frac{V_a}{nV_T}\right)}{I_{Load} / \left(I_S \frac{W}{L} e^{\frac{V_C}{nV_T}} \right) + 1} \right). \quad (27)$$

However, achieving an appropriate compensation voltage with minimal area and circuit overhead remains one of the practical challenges in system implementation. To address this issue, researchers have proposed various innovative compensation schemes. For example, the multi-stage passive compensation bridging method introduced in [29] automatically generates the required compensation voltage through inter-stage gate coupling, thereby eliminating the need for external

TABLE I

COMPARISON BETWEEN THRESHOLD-VOLTAGE COMPENSATION AND LEAKAGE SUPPRESSION TECHNIQUES

Technique Category	Primary Mechanism	Key Performance Impact	Target Input Condition
Threshold-Voltage Compensation (V_{TH})	Overcomes intrinsic threshold voltage limitations to facilitate transistor conduction.	Higher Sensitivity: Enables the front-end to cold-start and generate sufficient supply voltage to activate subsequent circuits.	Ultra-low input power (Weak ambient signals)
Leakage Suppression	Reduces energy loss caused by reverse leakage currents, which dominate at elevated voltages.	Extended PDR & Higher PCE: Enhances overall efficiency and maintains system stability across a wider range of fluctuating inputs.	Wide/high input power (Strong/complex ambient signals)

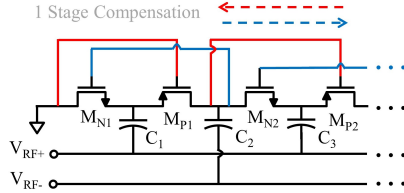


Fig. 8. Principle of multi-stage passive compensation topology in Dickson rectifier.

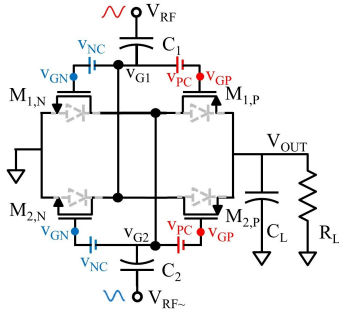


Fig. 9. Principle of CCDD rectifier with gate-biased VTH compensation technology.

bias circuitry. This approach also allows further performance optimization by adjusting the number of bridged stages, as illustrated in Fig. 8. Other works employ auxiliary transistors to modify the gate-connection topology of the conventional Dickson rectifier, thereby introducing threshold-voltage compensation [28], [32]. For example, authors in [32] utilized floating-gate transistors to provide a high-resistance bias path for establishing the required DC bias, while [28] derives the optimal compensation voltage and dynamically generates it using a subthreshold auxiliary transistor. This approach overcomes the drawback of traditional techniques, in which the compensation voltage varies with the input power level.

Similar design strategies have also been adopted in CCDD rectifier implementations, where threshold-voltage compensation is employed to enhance overall performance. To further guide the design of CCDD rectifiers with V_{TH} compensation, we develop a model for the CCDD structure incorporating additional gate biasing. The compensated CCDD rectifier can be equivalently represented by the circuit shown in Fig. 9, in which two DC bias voltages, V_{NC} and V_{PC} , are added on top of the original gate voltage V_G . Consequently, the subthreshold

currents of $M_{1,N}$ and $M_{2,P}$ can be expressed as:

$$I_{Dsub,M_{1,N}} = I_{S,N} \frac{W_N}{L_N} \left(e^{\frac{V_G + V_{NC} + V_{RF} \cdot \sin \omega t}{V_T}} - e^{\frac{V_{NC} + 2V_{RF}}{V_T}} \right) \times (1 + \lambda_{sub}(V_G - V_{RF} \cdot \sin \omega t)). \quad (28)$$

$$I_{Dsub,M_{2,P}} = I_{S,P} \frac{W_P}{L_P} \left(e^{\frac{[V_{OUT} - (V_G + V_{PC})] + V_{RF} \cdot \sin \omega t}{V_T}} - e^{-\frac{V_{PC} + 2V_{RF}}{V_T}} \right) \times (1 + \lambda_{sub}((V_{OUT} - V_G) - V_{RF} \cdot \sin \omega t)). \quad (29)$$

where V_{NC} and V_{PC} represent the effective static DC compensation voltages applied to the NMOS and PMOS devices, respectively. Conceptually, these constant DC voltage offsets shift the instantaneous V_{GS} , thereby lowering the apparent threshold voltage barrier of the CMOS devices to significantly enhance forward charging current under ambient weak-signal conditions. Using the derivation presented in Section II, the expressions for the gate voltage V_G and the output voltage V_{OUT} can be obtained as follows:

$$V_G = V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_N e^{\frac{V_{NC}}{V_T}} \cdot I_0 \left(\frac{2V_{RF}}{V_T} \right)}{K_N e^{\frac{V_{NC}}{V_T}} \cdot I_0 \left(\frac{V_{RF}}{V_T} \right)} \right). \quad (30)$$

$$V_{OUT} = V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_N e^{\frac{V_{NC}}{V_T}} \cdot I_0 \left(\frac{2V_{RF}}{V_T} \right)}{K_N e^{\frac{V_{NC}}{V_T}} \cdot I_0 \left(\frac{V_{RF}}{V_T} \right)} \right) + V_T \ln \left(\frac{\frac{I_{Load}}{2} + K_P e^{\frac{V_{PC}}{V_T}} \cdot I_0 \left(\frac{2V_{RF}}{V_T} \right)}{K_P e^{\frac{V_{PC}}{V_T}} \cdot I_0 \left(\frac{V_{RF}}{V_T} \right)} \right). \quad (31)$$

Therefore, the applied compensation bias directly influences the output voltage of the rectifier. It is also worth noting that the electron mobility in NMOS channels is significantly higher than the hole mobility in PMOS devices. In addition, NMOS transistors in practical designs can often be replaced with depletion-type devices that feature an even lower threshold voltage. As a result, for the same width-length ratio, the charging speed of an NMOS transistor is much faster than that of a PMOS transistor. This discrepancy typically forces the PMOS devices to adopt a much larger width-length ratio to achieve optimal rectification performance, leading to considerable area overhead.

With compensation, however, the term $K \cdot e^{\frac{V_C}{V_T}}$ can be treated as a whole. By providing appropriate compensation

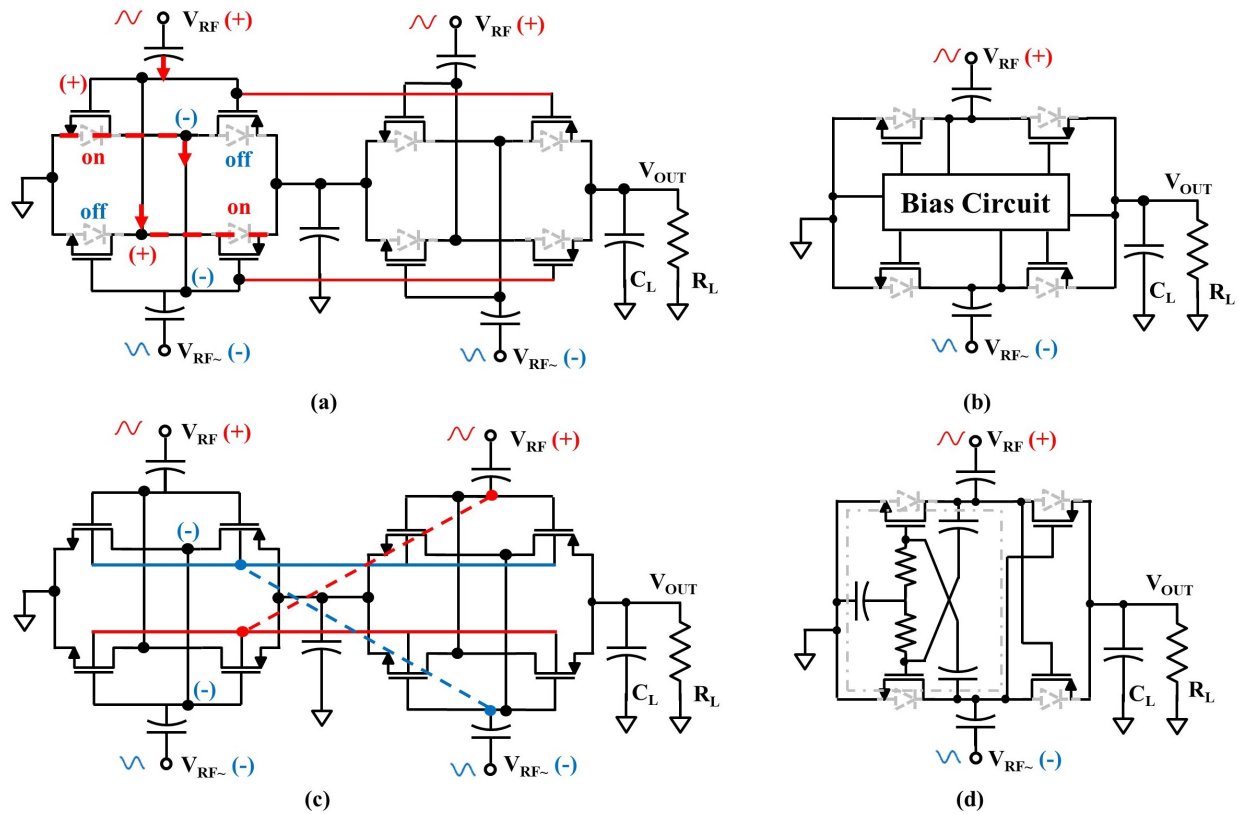


Fig. 10. Prior art techniques for threshold voltage compensation in CCDD rectifier. (a) Compensation through cross-stage connection for PMOS's gate [17], [25]. (b) Compensation through auxiliary bias circuit [33]. (c) Compensation through cross-stage connection for transistors' body [34]. (d) Compensation through active bias tuning [35].

for the PMOS transistors, their effective charging speed can be increased without resorting to excessively large device dimensions. This enables higher sensitivity and efficiency with substantially reduced area cost [25].

The above discussion summarizes the basic principles of threshold-voltage compensation in CCDD rectifiers. In practical circuit implementations, designers adopt different techniques to realize such compensation [17], [25], [33]–[35], as illustrated in Fig. 10. Based on the location of the compensation, these methods can be categorized into gate-bias compensation and body-bias compensation. The former directly modifies the transistor's V_{GS} , whereas the latter reduces the threshold voltage by influencing V_{SB} . From the perspective of circuit architecture, compensation schemes can be further divided into self-compensation and compensation using additional bias circuitry. Self-compensation adjusts only the internal connection topology to achieve inter-stage compensation, whereas the latter employs dedicated auxiliary circuits to generate the required compensation voltage.

Furthermore, given the inherent advantages of NMOS devices in rectifier circuits, some studies attempt to employ an all-NMOS design, which effectively reduces the high threshold voltage associated with traditional PMOS transistors while eliminating their area overhead. Such designs can also be combined with additional techniques to further enhance performance [36]–[38]. For example, [36] introduces a hybrid CC–DB architecture built upon an all-NMOS transistor

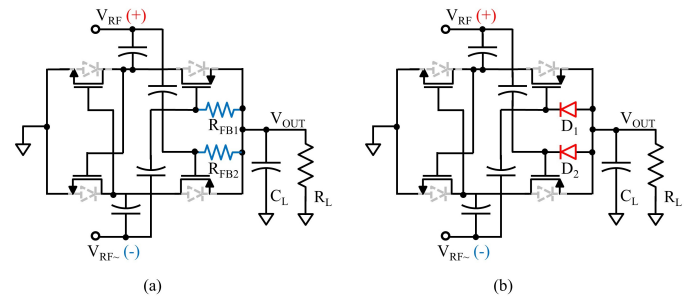


Fig. 11. Feedback-based leakage suppressing topologies [43], [44]. (a) Schematic of self-biased rectifier with feedback resistors. (b) Schematic of self-biased rectifier with feedback diodes.

structure, leveraging the strengths of both approaches and achieving an ultra-wide power dynamic range (PDR) of 22.3 dB. In [37], an internal threshold-voltage cancellation (IVC) technique is incorporated to obtain performance comparable to that of CCDD rectifiers while nearly eliminating reverse leakage current.

B. Leakage Suppressing Topologies

As discussed above, threshold-voltage compensation primarily addresses the conduction limitation encountered under ultra-low input conditions. Nevertheless, when sufficient RF energy becomes available and the rectifier output voltage increases, the dominant loss mechanism gradually shifts to re-

TABLE II
RECTIFIER PERFORMANCE COMPARISON

Technique	Literature	Architecture	PDR (dBm) (PCE > 30%)	Sensitivity (dBm) @ 1V	Peak PCE @ RF Input Power	Frequency	Chip Area (mm ²)	Technology
V_{TH} Compensation	A. K. Moghaddam [80] (2017)	Self-body-biasing	-5 to 5	-8 @ 50k Ω	73.9% @ 5 dBm	953 MHz, 2GHz	0.029	130 nm
	S. M. Noghabae [25] (2021)	Cross-connected gate biasing	-21.5 to -9.5	-29 @ 100M Ω	42.4% @ -16 dBm	915 MHz	0.029	130 nm
	M. A. Akram [81] (2023)	Bootstrap (BS)	-6 to (>10)	NA	68.5% @ 0 dBm	433.92 MHz	0.0035	180 nm
	H. Jiang [33] (2024)	Auxiliary self-bias	-24 to -12	-22.5 @ ∞	42.7% @ -20.9 dBm	915 MHz	0.0636	130 nm
Leakage Suppressing	M. H. Ouda [42] (2016)	Switch-based adaptive-bias	-27 to -11	-18 @ 100k Ω	65% @ -20.8 dBm	1 GHz	0.0085	180 nm
	M. H. Ouda [43] (2017)	Resistor-based feedback biasing network	-23 to (>-10)	-17 @ 100k Ω	65.4% @ -15.2 dBm	433 MHz	0.017	180 nm
	A. S. Almansouri [44] (2018)	Diode-based feedback biasing network	-30 to (>-12)	-19.2 @ 100k Ω	86% @ -19.6 dBm	433, 900 MHz	0.0084	180 nm
	X. Li [37] (2025)	Internal V_{TH} cancellation with All-NMOS	-9.5 to 4	-15.5 @ 1M Ω	55% @ -4.39 dBm	900 MHz	0.022	180 nm
	Y. Lee [40] (2025)	Switch-based biasing network	-22.5 to -5	-22 dBm @ 500k Ω	74.43% @ -18dBm	900 MHz	0.015	65 nm
	Y. Lee [39] (2025)	Switch-based self-adaptive biasing	-29 to -8	-18 dBm @ 100k Ω	84.3% @ -21dBm	900 MHz	0.0105	65 nm

verse charge leakage. Since MOS transistors inherently operate as bidirectional switches rather than ideal diodes, charge stored on the load capacitor may flow back to the input whenever the instantaneous input voltage falls below the output voltage.

This reverse leakage current becomes increasingly pronounced at elevated input-power levels and is further aggravated by subthreshold conduction and gate-tunneling effects in nanoscale CMOS technologies. Consequently, the achievable PCE deteriorates rapidly as the input power increases, thereby severely limiting the high-efficiency PDR of the rectifier. For this reason, leakage suppression has emerged as another fundamental design direction that complements threshold-voltage compensation and enables RF energy harvesters to maintain both high efficiency and stable operation over a wide range of practical input conditions.

The fundamental principle of leakage suppression is to maximize the effective threshold voltage of the rectifying transistors during the reverse-biased phase, thereby maintaining them in a deep cutoff state. This is typically achieved by reducing the gate-source voltage (V_{GS}) or dynamically increasing the intrinsic V_{TH} . To this end, early literature frequently adopted self-biasing generation techniques analogous to those used for threshold compensation. These methods utilize the rectifier's own DC output (V_{OUT}) or the input RF signal to supply dynamic bias voltages to the gate or body terminals of the transistors.

In [43] and [44], the authors incorporated two additional DC-blocking capacitors to decouple the DC potentials of the PMOS and NMOS gates, thereby facilitating independent optimization of their DC bias points. As illustrated in Fig. 11, feedback resistors and feedback diodes are employed, respectively, to establish the necessary gate biasing.

Specifically, the resistive feedback configuration pulls the PMOS gate potential toward V_{OUT} , ensuring that the DC V_{GS} approaches zero. Consequently, the transistors enter a deep cutoff region during the reverse-biased phase, substantially

mitigating reverse leakage current. However, this comes at the expense of a slight degradation in forward conduction capability.

In contrast, the diode-feedback architecture introduces an adaptive mechanism. Leveraging the unidirectional conductivity of the diode, the feedback path effectively presents an open circuit under low-input conditions, maintaining an operation mode identical to that of a conventional CCDD rectifier. Conversely, under high-input scenarios, the feedback diode conducts, connecting the PMOS gate to the output voltage. This action reduces the effective source-gate voltage $|V_{SG}|$, thereby significantly suppressing reverse leakage without compromising sensitivity. As a result, the high-efficiency range of the rectifier is successfully extended. Nevertheless, the intrinsic threshold voltage of the feedback diode introduces a voltage drop within the feedback loop, rendering its leakage suppression capability slightly inferior to that of the resistive approach.

To mitigate the voltage drop inherent in diode-based feedback, subsequent research has pivoted toward switch-based low-loss biasing architectures. In [42], the authors connected the PMOS gates to the output node via a switch MOS transistor governed by a voltage monitor. This approach reduces the threshold voltage drop along the feedback path under high-input conditions while preserving sensitivity; however, the power consumption of the active monitor remains a non-negligible constraint. A comparable strategy is presented in [40], which introduces a switching network directly controlled by the RF input signal that activates exclusively during high-voltage states. Experimental results demonstrate that, operating within the GSM900 band, this technique achieves a PDR of 22 dB and a peak PCE of 74.43%, with a compact chip area of only 0.0154 mm².

Analyzing the aforementioned leakage suppression schemes reveals a common underlying principle: they utilize various feedback and biasing techniques to couple the output voltage

to the PMOS gates, thereby reducing $|V_{GS}|$ to suppress leakage. However, this mechanism often inadvertently compromises the forward charging capability. Consequently, to further enhance performance, [39] proposed a “switch-as-resistor” concept. By utilizing a V_{RF} -controlled inverter to generate timing-control signals, this design ensures that the equivalent feedback resistance of the NMOS switch appears solely during the leakage phase and vanishes during forward conduction. This topology achieved a dynamic range of 25 dB and a peak PCE of 84.3%.

Table II summarizes and compares the performance metrics of rectifiers reported in representative works [25], [33], [37], [39], [40], [42]–[44], [80], [81]. Since the output voltage is strongly dependent on the input power and load conditions, sensitivity is adopted as the primary normalized comparison metric. Based on their primary design objectives, these techniques can be broadly categorized into two groups: threshold voltage compensation and leakage suppression. From a practical design perspective, these techniques target different application requirements. For ultra-low-power IoT sensor nodes operating under extremely weak ambient RF conditions, achieving reliable cold-start capability and maximizing sensitivity are usually the primary concerns, making threshold-voltage compensation the preferred choice. Conversely, wearable and biomedical applications often require stable and efficient operation over dynamically varying input-power conditions, where leakage suppression techniques become more attractive due to their wider high-efficiency PDR. Nevertheless, these two approaches are not fundamentally contradictory, and recent reconfigurable rectifier architectures increasingly integrate both techniques to accommodate broader application scenarios.

IV. SYSTEM-LEVEL RFEH DESIGN FOR HIGH SENSITIVITY, LONG-TERM STABILITY, AND FUTURE TRENDS

Sections II and III establish the rectifier-level foundation of RFEH by analyzing how threshold-voltage compensation and leakage suppression improve weak-input sensitivity and extend the high-efficiency input-power range. However, successful cold start and sustained sensor operation cannot be ensured by rectifier optimization alone, because the usable energy is further determined by the temporal and spectral availability of RF power, the interaction between the rectifier and power-management unit (PMU), energy buffering and regulation, and the operating behavior of the load. Accordingly, this section shifts the discussion from rectifier-level optimization to system-level RFEH design, focusing on multiband energy aggregation, long-term stable power delivery, wide-PDR architectures, practical implementation considerations, and future research directions.

A. Multi-Band Hybrid RFEH System

High sensitivity denotes the capability of a circuit to generate a supply voltage sufficient for low-power systems (typically 0.9 V to 1.5 V) from extremely low input power levels. Consequently, in performance benchmarking, sensitivity is frequently defined as the minimum input power required to

TABLE III
REPORTED AMBIENT RF POWER LEVELS ACROSS REPRESENTATIVE LOCATIONS AND FREQUENCY BANDS.

Location	Measured RF Power (dBm)				Reference
	DTV	GSM850 & GSM900	GSM1800 & GSM1900	Wi-Fi	
Underground stations	-75	-37	-40	-70	[45]
Residential areas	-47.05	-51.92	-54.52	-67.165	[46]
Metro stations	-43.12	-40.30	-41.65	-65.615	[46]

sustain a 1 V output. However, the output voltage amplitude is governed not only by the input power and rectifier efficiency but also significantly by the loading conditions. For instance, given a typical ambient input of -20 dBm ($10 \mu\text{W}$), even a theoretical RF frontend with 100% Power Conversion Efficiency (PCE) could only drive a $100 \text{ k}\Omega$ resistive load at 1 V. This power budget is evidently inadequate for the energy demands of most systems equipped with wireless communication capabilities. Therefore, to ensure stable energy harvesting in complex environments while aggregating more power sources, the research focus has extended to multi-band RF energy harvesting systems.

The measurements summarized in Table III illustrate the substantial variation in ambient RF power across frequency bands and deployment environments [45], [46]. Owing to their superior power availability, the GSM850 and GSM900 bands have rendered the 900 MHz frequency the predominant choice for the majority of RF energy harvesting system designs [17], [25], [37], [40], [47], [48]. Nevertheless, due to the inherent spatiotemporal fluctuations of RF energy, the dominant frequency band varies significantly depending on the specific region, and the potential energy contributions from other bands cannot be disregarded [82], [83]. For instance, substantial energy can often be harvested from 2.4 GHz or 5 GHz Wi-Fi signals in indoor environments [27], [49].

Consequently, developing strategies to effectively harvest energy from multiple frequency bands has emerged as a primary concern for designers. In [57], the authors presented a dual-band energy harvesting system capable of switching between Wi-Fi 6 (5 GHz) and Wi-Fi 6E (6 GHz) by actively tuning the capacitance within the matching network. Conversely, to achieve multi-band operation, the work in [55] utilized transmission lines to construct a complex impedance matching network. By optimizing the microstrip line parameters, this design achieved impedance matching at three distinct frequency points ($S_{11} < -14$ dB). However, the stringent matching requirements imposed significant design challenges, rendering the system highly sensitive to PCB layout variations and substrate parameters. Furthermore, while the aforementioned works extend frequency coverage and enhance system robustness, they typically operate by harvesting from only a single band at any given instant. As a result, they fail to aggregate power from multiple sources simultaneously,

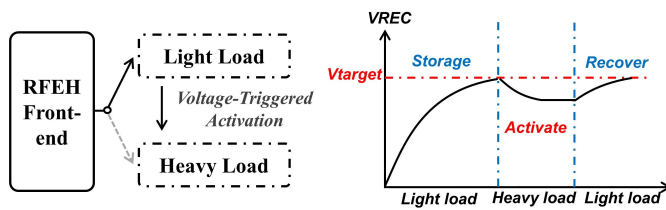


Fig. 12. Duty-cycle high-sensitivity RFEH system.

limiting the total harvested energy.

To further maximize the total harvested power, the efficient aggregation of energy from multiple frequency bands has emerged as a critical challenge. In [21], the authors proposed a dual-band RF energy harvesting system integrated with a wideband antenna. This system captures signals at 900 MHz and 1900 MHz, utilizing two parallel RLC networks to achieve frequency isolation. The signals are then directed to separate rectifier modules to generate dual outputs, thereby realizing concurrent energy extraction at the RF front-end. Conversely, the work in [84] focused on the post-rectification stage, merging the dual DC outputs generated by multi-band rectifiers to achieve superior sensitivity.

These designs demonstrate the feasibility of multi-band energy harvesting and provide a theoretical foundation for scaling to additional frequency bands. Furthermore, extensive research is currently exploring hybrid energy harvesting from heterogeneous sources. This approach aims to maximize power acquisition while mitigating the inherent limitations of single-source systems, such as the environmental dependency of solar energy and the low power density of RF signals [59]–[61].

B. High-Sensitivity High-Efficiency RFEH System

Despite the fact that multi-band RFEH can supply more energy to the system, achieving ultra-high sensitivity, defined as the harvester's capability to cold-start under extremely low input power, remains a paramount objective for designers. Over the past decades, research has focused primarily on improving rectifier performance by employing various compensation techniques to reduce the required input amplitude and by cascading multiple stages to boost the output voltage. For example, [25] achieves a 1-V output under a -29 dBm input and a $100\text{ M}\Omega$ load using dual compensation and a 10-stage rectifier. In [63], the authors further push sensitivity to -36.5 dBm by adopting a 108-stage cascade at the cost of significantly increased area and reduced efficiency.

It is worth noting that such ultra-high sensitivities are typically demonstrated under high-impedance or even capacitive load conditions. This raises an important concern: although the power consumption of integrated circuits continues to decrease, sensor nodes powered by RF energy often require wireless communication capability, which in turn depends on power-hungry clock-generation modules (oscillators or PLLs) and transmitters. Their power consumption is generally at the milliwatt level. Assuming a 1-V operating voltage, the equivalent load resistance during activation is only on the order of tens to hundreds of kilo-ohms—far from what can be driven in real time under ultra-low RF input power.

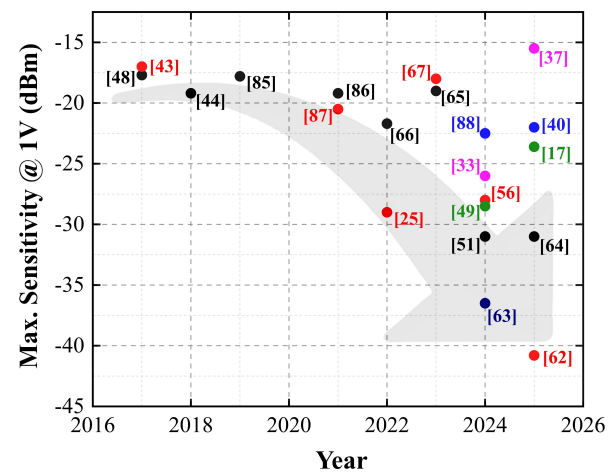


Fig. 13. Sensitivity improvement in RFEH systems.

Therefore, to further enhance both the sensitivity and load-driving capability of RF energy harvesters, a large energy storage capacitor along with a switch network is typically inserted between the rectifier and the load. The system alternates between charging and discharging phases, as illustrated in Fig. 12. During the charging phase, the switch remains open, and the rectifier effectively sees a capacitive load, allowing the storage capacitor to be charged to a high voltage. Once the voltage reaches the operating threshold, the switch closes, the capacitor discharges into the load, and the load performs its task. As the voltage continues to drop, the system eventually becomes unable to support normal operation, and the switch opens again, waiting for the next activation cycle.

Within this operating framework, only ultra-low-power voltage-monitoring circuits are active during the charging phase, resulting in a high effective load impedance that enables operation under weak input conditions. Similar concepts are demonstrated in [62] and [51]. In [62], the system achieves a 0.87 V output at an input power as low as -40.8 dBm and is capable of harvesting $8.7\text{ }\mu\text{J}$ from a -35.2 dBm ambient RF signal within 1.6 hours, providing a battery-free solution for IoT devices. In [51], the system achieves a 20-s cold-start wake-up time under a -31 dBm input.

In addition, sensitivity can also be improved by interfacing the rectifier with a DC–DC converter. For instance, [64] combines a three-stage subthreshold-biased rectifier with a maximum-power-point-tracking (MPPT)-enabled charge pump and proposes a burst-charging scheme, achieving a sensitivity of -31 dBm while maintaining high efficiency.

Fig. 13 summarizes the sensitivity improvements achieved by recently reported RFEH systems [17], [25], [33], [37], [40], [43], [44], [48], [49], [51], [56], [62]–[67], [85]–[88]. With continuous advancements in circuit techniques, system-level sensitivity has improved rapidly, and such designs may soon become feasible for real-world battery-free sensor nodes.

C. Wide-PDR High-Stability RFEH System

With sufficient input energy and high sensitivity achieved, RFEH systems must further address various unexpected conditions present in complex real-world environments to ensure

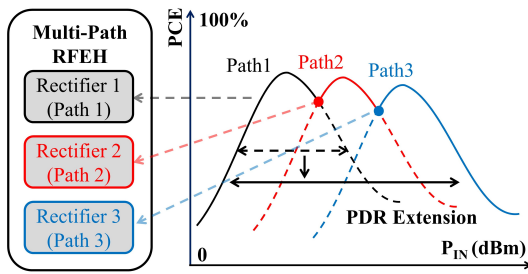


Fig. 14. PCE combination and PDR extension using a multipath rectifier.

compatibility with diverse battery-free applications. This requires both a wide high-efficiency power conversion range and stable output regulation. The former ensures high energy utilization across both low- and high-input-power scenarios, thereby supplying adequate energy to subsequent functional circuits. The latter is motivated by both application requirements and the inherent characteristics of RFEH systems. Due to the limited drive capability of RF harvesters, the connected load circuits typically require extremely low static power consumption, which relies on numerous switches and transistors operating in the subthreshold region. As a result, these circuits become highly sensitive to process, voltage, and temperature (PVT) variations. Therefore, we expect the rectifier to maintain stable output voltage while achieving high energy-conversion efficiency over a wide input range.

At the circuit level, leakage suppression techniques can help achieve a wider PDR. At the system level, researchers have proposed various architectures, with one common approach being switching between different rectifier modes. For example, a system may operate in the CCDD-rectifier mode under low input power, and switch to a Dickson rectifier at high input power, thereby exploiting the strengths of both structures and maintaining high energy-conversion efficiency across wide input conditions [65], [67]. Other systems employ multiple rectifier paths tailored to different input levels, effectively stacking their peak-efficiency points [48], as illustrated in Fig. 14.

Some studies further reconfigure the number of rectifier stages to adjust the effective input impedance under different input energy conditions [17], [18], [47], [68]. This improves impedance matching over a wider range, significantly reducing power loss between the antenna and rectifier and enhancing overall power-transfer efficiency.

It is also important to note that the number of rectifier stages directly affects the output voltage. Several works exploit this characteristic by reconfiguring stage count to regulate the output voltage. This not only enables high efficiency and a wide PDR but also ensures stable and controllable output voltage over a broad input range, thereby improving overall system robustness [17]. Additionally, the periodic operation scheme discussed in Section IV.B can further enhance system stability by constraining the operating voltage within predefined limits, ensuring that the functional circuits reliably start up when the switch turns on. However, when designing duty-cycled RFEH systems, the storage capacitor size and reference

voltage must be carefully adjusted based on the characteristics of the connected load to prevent the system from losing power before completing its intended operation.

D. Future Trends and Challenges

Undoubtedly, achieving an RFEH front end with high sensitivity, high efficiency, wide PDR, and stable output has always been a primary objective for designers. Such performance must be realized through joint optimization at both the circuit and system levels. Observing the literature in recent years, an increasing number of works adopt an application-driven perspective, where functional circuits, power management units, and RFEH front ends are co-designed to form a complete closed-loop system [72], [73], [75].

Within such a system, each module introduces its own specialized techniques, such as event-driven architectures, UWB signaling, time-division multiplexing, and frequency-division multiplexing in low-power functional circuits [76], [77]; MPPT, power gating, adaptive biasing, and pA-level zero-bias leakage-based bias generation in power management modules [2]–[4], [49], [75], [78]; and various compensation, leakage suppression, reconfiguration, and mode-switching strategies in RFEH front ends. When integrated together, these modules often create new design opportunities and enable the practical deployment of RF-enabled battery-free systems.

Among these developments, the practical deployment of RFEH systems faces significant physical and system-level challenges. At the physical implementation level, layout-dependent parasitic effects, such as the Well Proximity Effect (WPE) and Shallow Trench Isolation (STI) stress, can alter intrinsic threshold voltages, while grounding the bulk of native transistors in multi-stage rectifiers introduces complex body effects. Furthermore, to mitigate phase mismatches induced by process variations, designers must frequently employ strategic dummy devices, common-centroid layouts, and parallel unit-sized capacitor arrays. Beyond these physical non-idealities, the system-level combination of RFEH front ends with ultra-low-power power management units represents another immediate challenge.

Duty-cycled operation offers a viable approach for cold-starting under weak input power. However, this scheme imposes stringent requirements on the PMU and voltage-detection circuitry, necessitating strictly minimized quiescent power during the charging phase and rapid response upon activation, which inherently introduces stability concerns.

In addition, the co-design of the rectifier, antenna, and matching network is becoming an important trend. The passive components and antennas that constitute off-chip matching networks typically occupy areas much larger than the chip itself, making fully integrated solutions difficult, particularly for implantable medical applications. To date, only a limited number of works have demonstrated complete RFEH front ends with fully integrated matching networks.

Finally, the rapid development of artificial intelligence (AI) and machine learning (ML) has introduced new opportunities for next-generation RF energy harvesting systems. AI-assisted approaches may enable adaptive impedance match-

ing, dynamic rectifier reconfiguration, and intelligent duty-cycle control according to both the temporal characteristics of ambient RF energy and the sensing requirements of target applications.

Nevertheless, directly integrating sophisticated AI inference engines into self-powered RFEH systems remains highly challenging because the power budgets of battery-free sensor nodes are typically limited to the nW– μ W range, which is substantially lower than the power consumption of most contemporary ML accelerators. Therefore, a more practical near-term solution may be to employ AI/ML algorithms at external infrastructure nodes, such as edge servers or RF base stations, and feed lightweight control signals back to the RFEH system for adaptive operation. As ultra-low-power AI hardware continues to evolve, limited forms of on-chip intelligence may gradually become feasible, opening new possibilities for intelligent and self-sustainable RFEH systems.

V. CONCLUSION

This manuscript provides a comprehensive overview of RFEH front-end design and implementation for low-power devices, emphasizing sensitivity enhancement, leakage reduction, and stable operation across wide input conditions. Through modeling of CCDD rectifier behavior under low RF input, we reveal the key factors influencing charge transfer and efficiency. Existing compensation structures, reconfigurable rectifiers, and adaptive matching techniques were reviewed and discussed to illustrate their strengths and limitations. At the system level, strategies for improving robustness under dynamic environments were summarized. Looking ahead, integrating intelligent control, wide-range impedance adaptation, and low-leakage architectures will be essential for achieving high-efficiency, battery-less operation in future IoT and biomedical applications.

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